Zirui Zhou

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EDUCATION

Ph.D. **Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University** Starting Sept 2027
- Incoming Ph.D. Student
- **Supervisor:** Tao Du

B.S. **Department of Automation, Tsinghua University** Sept 2023 - Current
- **Courses:** Statistics, Linear Algebra, Calculus, Data Structures and Algorithms, Machine Learning, etc.

H.S. **The High School Affiliated to Renmin University of China, Beijing, China** Sept 2020 - Jul 2023

AWARDS AND HONORS

Second Prize, 44th Challenge Cup, Tsinghua University 2026
- A system for 3D scene change modeling and continual reconstruction with consistent multi-view aggregation
- **Supervisor:** Songchun Zhu

Science and Technology Innovation Excellence Scholarship, Tsinghua University 2024-2025

Academic Progress Scholarship, Tsinghua University 2024-2025

Grand Prize (Top Award), 7th Software Design Competition, Tsinghua University 2024
- Development of a music puzzle game using Unity
- Link to my project: https://github.com/zr-zhou0o0/light

Grand Prize (Top Award), 3rd Robot Dog Development Competition, Tsinghua University 2024
- Robot dog soccer matches
- Link to my project: https://github.com/zr-zhou0o0/Cyberdog-7-3

Science and Technology Innovation Excellence Scholarship, Tsinghua University 2023-2024

Arts and Culture Excellence Scholarship, Tsinghua University 2023-2024

EXPERIENCES

Intern, Beijing Institute for General Artificial Intelligence Sept 2024 - Current
- **Research topic:** 3D scene decomposition and reconstruction, with hands-on experience in scene representations (SDF, NeRF, 3DGS, Voxel) and 3D generative models (TRELLIS, SAM3D).
- **Supervisor:** Yixin Chen

Algorithm Intern, Foundation Model Group, Galbot Jan 2026 – Aug 2026
- **Research topic:** Vision-Language-Navigation models, world models
- **Supervisors:** He Wang, Zhizheng Zhang

Intern, Graphics and Geometric Computing Group Apr 2024 - Aug 2024
Department of Computer Science and Technology, Tsinghua University
- **Research topic:** 3D scene layout generation
- **Supervisor:** Song-Hai Zhang

PUBLICATIONS

VLMs are Native Action Models: Unleashing the Manipulation Capabilities of VLMs with Explicit Action-Space Grounding
Jingkun Yi*, Shujie Zhang*, **Zirui Zhou**, Haoxiao Wang, Haiyang Sun, Yuhang Dong, Xudong Xu, Xing Shen, Tao Zhang, Yong Liu, Jiangning Zhang
International Conference on Learning Representations (ICLR), 2027 (Under review)
WebPage: https://general-robo.github.io/EASG/

3D Scene Change Modeling with Consistent Multi-View Aggregation
Zirui Zhou, Junfeng Ni, Shujie Zhang, Yixin Chen, Siyuan Huang
International Conference on 3D Vision (3DV), 2026
WebPage: https://zr-zhou0o0.github.io/SCaR3D/

DecomVoxel: Harnessing 3D-Native Priors with Guided In-situ Denoising Optimization for Decompositional Scene Reconstruction
Junfeng Ni*, **Zirui Zhou***, Yixin Chen, Yu Liu, Nan Jiang, Zhifei Yang, Songchun Zhu, Siyuan Huang
ACM Transactions on Graphics (TOG), SIGGRAPH Asia, 2026
WebPage: https://decomvoxel.github.io/DecomVoxel-Webpage/

RoboSnap: Sim-Ready Real-to-Sim Scenes for Generalizable Robot Learning and Evaluation
Shujie Zhang*, Jingkun Yi*, Weipeng Zhong, **Zirui Zhou**, Yangkun Zhu, Hanqing Wang, Xudong Xu, Weinan Zhang, Chunhua Shen
Conference on Robot Learning (CoRL), 2026
WebPage: https://robosnap.github.io/

Decompositional Neural Scene Reconstruction with Generative Diffusion Prior
Junfeng Ni, Yu Liu, Ruijie Lu, **Zirui Zhou**, Song-Chun Zhu, Yixin Chen, Siyuan Huang
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025
WebPage: https://dp-recon.github.io/

Synthesizing 3D Scenes via Diffusion Model that Incorporates Indoor Scene Characteristics
Liang Yue, Shao-Kui Zhang, Lin Yuan, YiTao Chen, **Zirui Zhou**, Song-Hai Zhang
Proceedings of the ACM International Conference on Multimedia (ACMMM), 2025

COURSE PROJECTS

Cross-Domain EgoCross Challenge @ EgoVis, CVPR 2026 Mar 2026 – May 2026
- Ranked 7/92 in the EgoCross challenge at EgoVis, CVPR 2026 by fine-tuning Vision-Language Models (VLMs) with Supervised Fine-Tuning (SFT) and LoRA for cross-domain egocentric video understanding.

Intelligent Voice Dialogue System Nov 2025 – Dec 2025
- Built a streaming voice dialogue system using FunASR, DeepSeek, and CosyVoice, implementing RAG-based memory search and voice style control.

Counting Game with Arduino Manipulator Aug 2025 – Sep 2025
- Integrated MediaPipe-based hand gesture recognition, serial communication, manipulator motion control, and game logic modules.

Imperial College London | Data Science and AI Summer Camp Jul 2025 – Aug 2025
- Developed a multi-layer brain slice segmentation model based on the U-Net architecture using PyTorch for automatic segmentation of different tissue layers in brain slice images, improving segmentation accuracy and efficiency through architecture and parameter optimization.

SKILLS

Tools: Git, LaTeX, Linux, Anaconda

Programming Languages: Python, C, C++, C#, HTML, ROS2

3D Modeling and Design
- Unity, Blender, Adobe Photoshop, Adobe Premiere
- Created visual art content with over 500,000 total views on social media

ACTIVITIES AND SOCIAL WORK

Orienteering Competition of John Mo Cup, Champion Apr 2024

Piano Performance: Merry Christmas, Mr. Lawrence Apr 2024
Student Festival of Department of Automation

"A-Star Exploration" Social Practice, First Prize Jan 2024

Tsinghua University Student Art Troupe Sept 2023 - Sept 2024

Middle-Distance Running Team of Department of Automation Aug 2023 - Mar 2024